Notes: Terviö (AER, 2008)

The Difference That CEOs Make: An Assignment Model Approach

Contents

1 Big picture

- **Question.** Can the very high and highly skewed levels of CEO pay be explained by competition for scarce managerial talent, and how much economic value do differences in CEO ability actually create?
- **Core idea.** CEO compensation and firm market values can be viewed as the competitive equilibrium of a two-sided assignment market: heterogeneous managers match with heterogeneous firms.
- **Two-sided heterogeneity.** The model allows both CEOs and firms to have fixed, indivisible quality differences. This differs from one-sided talent models in which firm size is entirely generated by managerial ability.
- **Complementarity.** Managerial ability is more valuable at larger firms because the same managerial skill affects more resources, assets, and future surplus. This creates positive assortative matching: better CEOs match with larger firms.
- **Method.** The paper uses observed CEO pay and firm market values to infer the unobserved distributions of CEO ability and firm-specific scale.
- **Empirical focus.** The calibration uses the 1,000 largest publicly traded U.S. firms from 1994 to 2004, with total compensation as CEO pay and Black–Scholes values for options.
- **Main finding.** Most variation in CEO pay is due to differences in firm characteristics, not large differences in CEO ability.
- **Economic magnitude.** Replacing all CEOs with the best observed CEO type would increase total surplus by only a few billion dollars across the 1,000 largest firms, a small amount relative to total market value.
- **Pay interpretation.** High CEO pay is consistent with competitive matching even when underlying ability differences are small, because small quality differences are magnified when matched to very large firms.
- **Why firm heterogeneity matters.** If firms were homogeneous, all CEO pay differences would have to be explained by talent. With heterogeneous firms, large pay differences can reflect Ricardian rents from firm scale.
- **Broader takeaway.** The assignment model explains how high CEO pay can arise in a competitive market while showing that the social value of marginal differences in CEO ability may be modest.

2 Incremental contribution of the paper

2.1 Contribution relative to classic scale-of-operations theories

- Earlier models by Mayer, Manne, Lucas, and Rosen emphasize that more talented managers control more resources and therefore earn more.
- In many classic models, firms are effectively homogeneous in fundamentals; firm size is an equilibrium consequence of allocating capital or labor to more talented managers.
- Terviö adds an explicitly two-sided assignment structure: firms themselves have fixed, exogenous differences in natural scale, brand, niche, technology, or sunk capital.
- This matters because observed firm size and CEO pay are not interpreted mechanically as evidence of large CEO talent differences.

Interpretation: The incremental contribution is to separate *firm scale rents* from *managerial ability rents*. This makes it possible to ask how much CEO ability matters after accounting for the fact that some firms are intrinsically more valuable places to manage.

2.2 Contribution relative to empirical executive compensation work

- Much empirical compensation work studies the incentive structure of CEO contracts: salary, bonus, options, and sensitivity to firm performance.
- Terviö instead focuses on pay *levels*: why CEOs of very large firms are paid so much and what those pay levels imply about ability differences.
- The paper links compensation levels to an equilibrium model rather than treating firm size as an ordinary regression control.
- It uses the joint distribution of pay and market value to recover the economic value of unobserved ability.

Interpretation: The paper turns the observed cross-section of CEO pay and firm value into an inference problem. It asks what latent talent distribution must rationalize observed pay in a competitive assignment equilibrium.

2.3 Contribution relative to the CEO pay debate

- One view interprets high CEO pay as evidence of rent extraction or weak governance.
- Another view interprets high pay as evidence of efficient competition for scarce talent.
- Terviö shows that high pay can be consistent with competitive assignment, but this does not imply that CEOs make enormous social contributions at the margin.
- The paper therefore distinguishes two questions:
  1. Can competitive markets generate high CEO pay? Yes.
  2. Are underlying ability differences large and socially important? The calibration suggests they are relatively small.

Interpretation: This distinction is the paper's most important conceptual advance. Competitive pay does not imply that the social product of the marginal CEO is huge; it may mostly reflect sorting across firms with very different natural scale.

2.4 Contribution to assignment-model methodology

- The paper derives equilibrium wage and profit profiles in quantile space.
- It shows how the slopes of observed factor-income profiles can be inverted to recover latent factor-quality profiles.
- It adapts the assignment model to long-lived firms by translating flow CEO pay and stock market values into comparable present-value objects.
- It handles adjustable capital by transforming observed market value into the part relevant for the assignment model.

Interpretation: Methodologically, the paper provides a way to infer unobserved talent and firm-quality distributions from observed pay and firm-value profiles.

3 Data and measurement

3.1 Sample

- The empirical analysis focuses on the 1,000 publicly traded U.S. firms with the largest market values in each year from 1994 to 2004.
- Firm market values are from Compustat.
- CEO pay is measured using total compensation, with option grants valued using standard Black–Scholes formulas.
- When a firm has multiple CEOs in a year, compensation is summed to measure the firm's total expected CEO compensation cost.

Interpretation: The sample is chosen to study the top end of the CEO market, where pay levels are most controversial and where the scale-of-operations effect should be strongest.

3.2 CEO pay as a flow and firm value as a stock

- CEO pay is observed as a flow compensation measure for the current year.
- Firm market value is a stock, capitalizing profits and future surplus.
- The model therefore needs assumptions that translate current pay into a present value of all future CEO payments.
- Under stationarity and constant growth, the present value of future CEO pay at firm i is proportional to current pay.

Interpretation: Without this translation, CEO pay and market value are not measured in common units. The paper's stationarity assumption makes the comparison tractable.

3.3 Exogenous firm scale

- The firm characteristic b is not observed directly.
- It represents exogenous firm-specific scale: niche size, brand value, technology, customer base, organizational capital, sunk capital, and other indivisible features.
- Observed market value is not equal to b because market value is partly an equilibrium outcome of the match with CEO ability.
- The paper recovers b indirectly from the pay and value profiles.

Interpretation: The distinction between observed market value and exogenous firm scale is central. Market value combines firm quality, CEO quality, and equilibrium rents.

3.4 Adjustable capital

Observed market value includes both:

$$v^* = v + k^*, \tag{1}$$

where v is the part of market value determined in the assignment problem and k^* is optimally chosen adjustable capital.

The paper transforms observed market value using

$$v_0 = \xi v_0^* - (1-\xi)\frac{w_0}{1-B}, \tag{2}$$

where the parameter ξ depends on the share of adjustable capital, the discount rate, and the growth-adjusted discount factor.

Interpretation: This adjustment prevents ordinary transferable capital from being mistaken for firm-specific scale in the assignment market.

3.5 Key calibration parameters

- r: discount rate.
- g: productivity or growth rate.
- $B = (1+g)/(1+r)$: growth-adjusted discount factor.
- θ: share of adjustable capital.
- λ: rate at which CEO impact fades over time.
- The paper reports results under several parameter scenarios to show robustness.

Interpretation: Because some parameters cannot be identified directly from the cross-section, the calibration reports a range of values rather than a single point estimate.

4 Assignment model

4.1 Factor qualities and matching

There is a unit mass of managers and firms. Manager ability is a and firm size or natural scale is b. The production function is

$$Y(a,b), \tag{3}$$

with

$$Y_a > 0, \qquad Y_b > 0, \qquad Y_{ab} > 0. \tag{4}$$

- $Y_a > 0$: higher ability increases output.
- $Y_b > 0$: larger firms produce more surplus.
- $Y_{ab} > 0$: ability is more valuable in larger firms.

Interpretation: The positive cross-partial generates positive assortative matching: the most able CEO is matched with the largest firm.

4.2 Quantile notation

Let $i \in [0,1]$ index the quantile of quality. The ability profile is

$$a[i] = a \quad \text{such that} \quad F_a(a) = i. \tag{5}$$

Similarly, the firm-quality profile is $b[i]$.

Interpretation: Quantile notation makes sorting transparent: manager i matches with firm i.

4.3 Competitive equilibrium conditions

The firm matched with manager i must prefer manager i to any other manager j at equilibrium wages:

$$Y(a[i], b[i]) - w[i] \geq Y(a[j], b[i]) - w[j]. \tag{6}$$

Participation constraints are

$$Y(a[i], b[i]) - w[i] \geq \pi^0, \tag{7}$$

$$w[i] \geq w^0. \tag{8}$$

The lowest active match breaks even:

$$Y(a[0], b[0]) = \pi^0 + w^0. \tag{9}$$

Interpretation: The equilibrium wage profile is pinned down by marginal sorting constraints and the participation constraint of the lowest-quality active CEO.

4.4 Wage and profit profiles

The marginal sorting condition implies

$$w'[i] = Y_a(a[i], b[i])a'[i]. \tag{10}$$

Integrating from the lowest active type gives

$$w[i] = w^0 + \int_0^i Y_a(a[j], b[j])a'[j]dj. \tag{11}$$

Analogously, the profit profile satisfies

$$\pi'[i] = Y_b(a[i], b[i])b'[i], \tag{12}$$

$$\pi[i] = \pi^0 + \int_0^i Y_b(a[j], b[j])b'[j]dj. \tag{13}$$

Interpretation: Factor incomes depend on the quality improvements below a given rank. Each side earns differential rents over lower-ranked competitors.

4.5 Multiplicative production

The empirical application uses

$$Y(a, b) = ab. \tag{14}$$

This functional form is not meant to assign literal output shares. It is a convenient normalization of two one-dimensional quality rankings.

Interpretation: Because a and b are ordinal quality indices, the model can normalize production multiplicatively without claiming that CEOs and firms receive Cobb–Douglas shares.

4.6 Scaling lemma

If the production function is scaled by G and outside opportunities scale by G, then equilibrium factor incomes scale by G:

$$Y_t(a, b) = GY(a, b), \qquad w_t^0 = Gw^0, \qquad \pi_t^0 = G\pi^0, \tag{15}$$

which implies

$$w_t[i] = Gw[i], \qquad \pi_t[i] = G\pi[i]. \tag{16}$$

Interpretation: Pure aggregate productivity growth scales CEO pay and firm profits without changing the shape of ability or firm-size distributions.

5 CEO-market adaptation

5.1 CEO impact over time

CEO ability can affect current and future surplus. Effective management ability at date t is

$$A_t = A(a_t, a_{t-1}, a_{t-2}, \ldots) = \sum_{\tau=0}^{\infty} \alpha_\tau a_{t-\tau}. \tag{17}$$

If CEO impact fades at rate λ, then the impact weights are

$$\alpha_\tau = \frac{\lambda}{(1+\lambda)^{\tau+1}}. \tag{18}$$

Interpretation: A CEO can affect future profits, but the impact decays over time. The parameter λ governs how quickly the impact fades.

5.2 Stationarity assumptions

The paper imposes three strong stationarity assumptions:

1. the ability distribution $a[i]$ and firm-size distribution $b[i]$ are constant over time;
2. productivity grows deterministically at rate g;
3. outside opportunities grow at rate g.

Interpretation: These assumptions allow the paper to map annual CEO pay into the present value of all CEO payments and to use market value as a comparable present-value object.

5.3 Present value of surplus

With current CEO type a at firm b, one-period surplus evolves as

$$y_t(a, b) = (1+g)^t ab. \tag{19}$$

The present value of surplus is

$$Y(a, b) = \sum_{t=0}^{\infty} B^t ab = \frac{ab}{1-B}, \tag{20}$$

where

$$B = \frac{1+g}{1+r}. \tag{21}$$

Interpretation: The assignment model is applied to present values, not just one-period flows.

5.4 Market value and CEO pay

The surplus at firm i is divided between future CEO pay and firm market value:

$$\frac{w[i]}{1-B} + v[i] = \frac{a[i]b[i]}{1-B}. \tag{22}$$

Interpretation: This equation is the key empirical bridge. It links observed CEO pay, observed firm value, and the latent product $a[i]b[i]$.

6 Inference of ability and firm-quality profiles

6.1 Differential-equation system

For the CEO market, the wage profile satisfies

$$w'[i] = \frac{\lambda}{\lambda + 1 - B} a'[i]b[i]. \tag{23}$$

Differentiating the present-value surplus relation and combining with the wage equation yields a second equation linking $a[i]$, $b[i]$, CEO pay, and firm value.

Interpretation: These two differential equations allow the paper to infer the relative shapes of the ability and firm-size profiles from observed compensation and market value.

6.2 Closed-form profiles

The relative ability profile is

$$\frac{a[i]}{a[0]} = \exp\left(\frac{\lambda}{\lambda + 1 - B} \int_0^i \frac{w'[j]}{w[j] + v[j](1 - B)} dj\right). \tag{24}$$

The relative firm-size profile is

$$\frac{b[i]}{b[0]} = \frac{w[i] + v[i](1 - B)}{w[0] + v[0](1 - B)} \cdot \frac{a[0]}{a[i]}. \tag{25}$$

Interpretation: The constants $a[0]$ and $b[0]$ are not separately identified, but relative abilities and relative firm qualities are identified.

6.3 Counterfactual value of ability

The model evaluates counterfactual matches. The key object is the surplus change from replacing the actual CEO at firm i with a CEO from quantile j:

$$\Delta Y(j, i) = Y(a[j], b[i]) - Y(a[i], b[i]). \tag{26}$$

Because the unknown scale constants cancel, these counterfactuals can be computed from observed profiles.

Interpretation: The calibration asks how much surplus would change if CEOs were rearranged or replaced by better or worse types.

7 Empirical design and identification

7.1 What is identified

- The model identifies relative CEO ability differences among top CEOs.
- It identifies relative firm-specific scale differences among large firms.
- It identifies counterfactual surplus changes from changing CEO-firm assignments.
- It does not identify absolute ability levels, only relative ability and economic impact within the top-firm sample.

Interpretation: The paper's estimates are structural implications of the competitive assignment model, not direct experimental effects.

7.2 Key identifying assumptions

- CEO pay and firm market values reflect competitive equilibrium prices.
- Matching is positive assortative.
- Ability and firm quality are one-dimensional.
- Firm and CEO quality distributions are smooth.
- The production function is multiplicatively separable in ability and firm size.
- Stationarity holds sufficiently to map pay flows and firm values into common units.

Interpretation: These assumptions are strong. The paper is best understood as asking what the competitive assignment model implies if taken seriously.

7.3 What the model abstracts from

- CEO effort and incentives are ignored.
- CEO bargaining power is ignored because the equilibrium is competitive.
- Governance frictions, board capture, and rent extraction are not modeled.
- The model does not explain why firms become large or why firm niches exist.
- The model does not explain the source of stock market booms, only their effect on pay through assignment.

Interpretation: The paper deliberately isolates the scale-of-operations mechanism and its implications for CEO ability valuation.

8 Tables and figures

8.1 Figure 1: comparative statics in the assignment model

Read:

- Figure 1 depicts the multiplicative production case $Y(a, b) = ab$.
- The increasing curve is the matching curve $a = \phi(b)$.
- Isoquants show output levels for different matches.
- The shaded regions illustrate how changes in factor-quality distributions change equilibrium wages and profits.

Economic message: The figure shows why factor owners are paid differential rents. In a complementary assignment model, changing the distribution of one side changes the rents of the other side through sorting.

8.2 Figure 2: CEO pay and firm rank in 2004

Read:

- Figure 2 plots CEO pay against firm rank by market value for 2004.

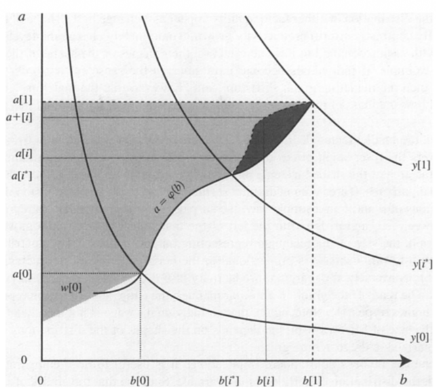

FIGURE 1. COMPARATIVE STATICS IN THE MULTIPLICATIVE CASE

Notes: The increasing curve covers both the active matches $\{b, \varphi(b)\}$ above $b[0]$ and potential but inactive matches below $b[0]$. The three decreasing curves are the isoquants for levels of output $y[0]$, $y[i^*]$, and $y[1]$. The entire shaded region is the equilibrium wage of the highest ability type, $a[1]$. The dark shaded region is the decrease in wage and increase in profits for the highest types if the matching curve between quantiles i^* and 1 were to shift up to the dashed line.

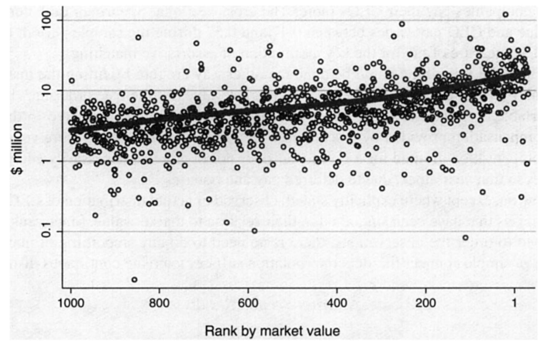

FIGURE 2. RELATION OF CEO PAY AND FIRM RANK BY MARKET VALUE IN 2004

- CEO pay rises strongly with firm rank, meaning larger firms pay more.
- The relationship is noisy but monotone enough to support the assignment calibration after smoothing.
- The slope of this relation is the empirical input that maps firm scale into CEO pay differentials.

Economic message: This figure is the central empirical motivation. The top firms pay much more, but the assignment model asks whether that reflects large talent differences or large firm-specific scale differences.

8.3 Table 1 and Figure 3: value of existing CEO ability

Read:

- Table 1 reports the value of CEO ability across the top 1,000 firms in 2004 under several parameter assumptions.
- The value over baseline is around \$21–25 billion depending on assumptions.
- The value of replacing all CEOs with the best observed type is only around \$3.2–3.4 billion.
- The value of replacing all CEOs with the median type is negative, around \$6.6–7.1 billion.
- Figure 3 plots the time series of the value of CEO ability and CEO rents relative to baseline ability.
- The value of ability rises around the late-1990s stock market boom and then falls after 2001.

Economic message: Ability differences matter, but their total value is modest relative to total market capitalization and CEO pay. The boom increases the value of ability because CEO decisions interact with future firm scale.

TABLE 1—VALUE OF EXISTING CEO ABILITY AT 1,000 LARGEST FIRMS (2004)
(*Values in $ billions*)

Assumptions	(A)	(a2)	(c)	(b2)	(B)		
Discount rate (r)	0.08	0.08	0.05	0.05	0.05		
Growth rate (g)	0.02	0.02	0.025	0.025	0.025		
Share of adj. capital (θ)	0	0.4	0.4	0.4	0.8		
Rate of impact fading (λ)[a]	∞	∞	0.5	0.1	0.1		
	Value of ability					Counterfactual rent to CEOs	
Results	(A)	(a2)	(c)	(b2)	(B)	(A)	(B)
Value over baseline	24.97	24.75	23.36	23.12	21.33	0	0
Value below max	3.16	3.17	3.25	3.27	3.40	2.85	3.36
Value over median	7.12	7.09	6.89	6.86	6.62	1.81	2.00
Actual totals							
CEO pay	7.12						
CEO rent[b]	4.39						
Market value	12,584.6						

Notes:
[a]Results for $\lambda = \infty$ calculated from the limiting values.
[b]Rent = pay − baseline pay.

Table 1: Value of existing CEO ability, 2004

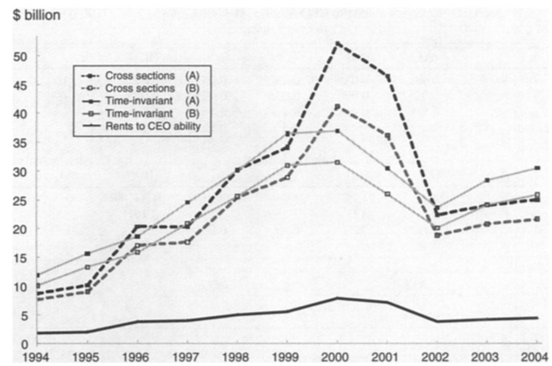

FIGURE 3. VALUE OF CEO ABILITY AND RENTS TO CEOS, RELATIVE TO BASELINE ABILITY, AT THE 1,000 LARGEST FIRMS

Notes: In "Cross Sections" the profiles of ability are inferred separately in each sample year, whereas in the "Time-Invariant" specification the distribution of ability is forced to be constant over time. The parameter assumptions with the most extreme results are shown for both specifications (see Table 1 for details). All values are in 2004 dollars.

Figure 3: Value of ability and rents over time

8.4 Figure 4 and Table 2: importance of firm heterogeneity

Read:

- Figure 4 evaluates counterfactual matches using three reference firms: the 1,000th firm, the 500th firm, and the largest firm.
- If all firms were as small as the 1,000th firm, even top CEO ability would add only a few million dollars.
- If the largest firm is the reference, lower-ranked CEOs would make much less surplus at that firm.
- Table 2 shows counterfactual CEO rents when firm size is homogenized.
- Counterfactual firm size changes dwarf the value of CEO ability changes.

Economic message: Firm heterogeneity is the main driver of pay differences. High pay at the top reflects the scale of the firm much more than large differences in CEO ability.

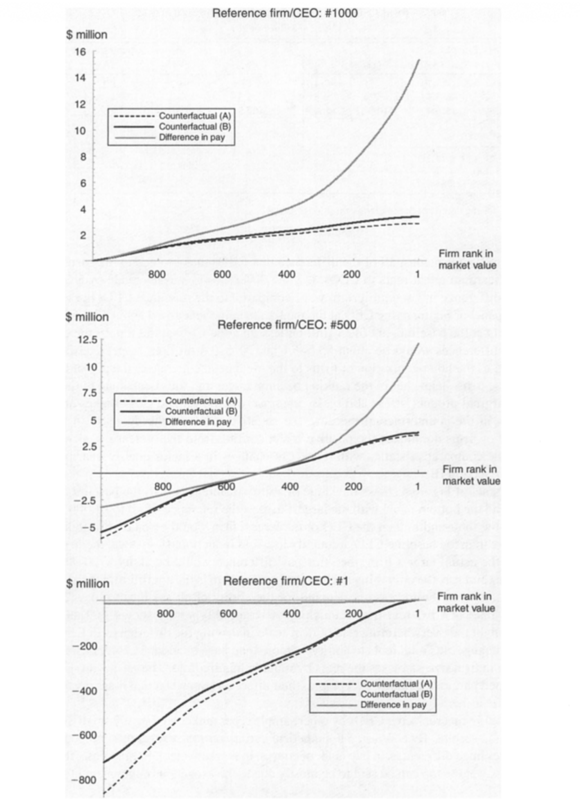

FIGURE 4. IMPACT OF CEO ABILITY BY REFERENCE FIRM

Notes: The counterfactual difference that CEOs would make to economic surplus created at the reference firm if they were to replace the actual CEO at the reference firm. The value is calculated under two parameter scenarios (see Table 1). The gray line depicts the difference in actual pay of the CEOs relative to that of the reference rank.

Figure 4: Impact of CEO ability by reference firm

TABLE 2—TOTAL CEO RENT UNDER COUNTERFACTUAL FIRM SIZE (2004)[a]

(All values in $ billions)

CEO pay over baseline pay						Counterfactual market value	
Reference firm	(A)	(a2)	(c)	(b2)	(B)	(A)	(B)
1,000th largest	1.70	1.71	1.78	1.79	1.92	1,221.9	6,673.6
500th	5.77	5.74	5.59	5.55	5.30	4,140.5	8,945.4
1st	516.4	510.3	469.3	462.5	413.3	370,663.3	108,255.2
Actual totals							
CEO rent	4.39						
Baseline pay×1,000	2.73						
Market value	12,584.6						

[a] See Table 1 for the details of parameter assumptions (A)–(B).

Table 2: CEO rent under counterfactual firm size

8.5 Figure 5: inferred CEO abilities over time

Read:

- Figure 5 plots inferred relative abilities at selected firm ranks from 1994 to 2004.
- The ability of CEOs at the top ranks is only modestly above the baseline 1,000th CEO.
- The largest inferred ability differences occur around the stock market boom.
- The average relative ability of the highest ranks is still close to one, not dramatically higher.

Economic message: The model infers small relative talent gaps. Competitive pay differences can be large even when ability gaps are small because large firms magnify small skill differences.

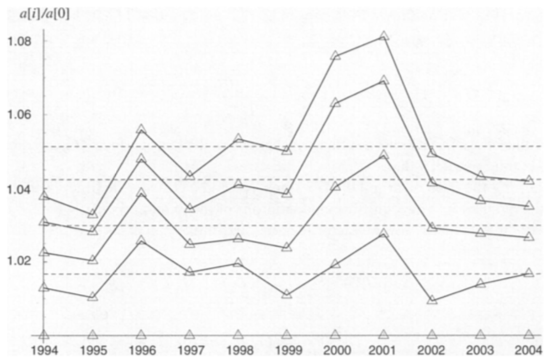

FIGURE 5. INFERRED CEO ABILITIES AT 1ST, 250TH, 500TH, AND 750TH LARGEST FIRM (RELATIVE TO 1,000TH) BY YEAR

Note: Dashed lines give the respective averages over this time period.

8.6 Figure 6 and Figure 7: time-invariant calibration

Read:

- Figure 6 compares actual CEO pay differences to model-predicted differences under a time-invariant ability and firm-size distribution.
- The model matches the broad rise in pay differences over time, especially for intermediate ranks.
- Figure 7 compares actual and predicted market values for selected ranks.
- The model's predicted market values capture broad movements but can overpredict values for smaller ranks in some years.

Economic message: A simple assignment model with time-invariant distributions can explain a meaningful part of the time-series movement in CEO pay through changes in aggregate scale, but it cannot explain everything.

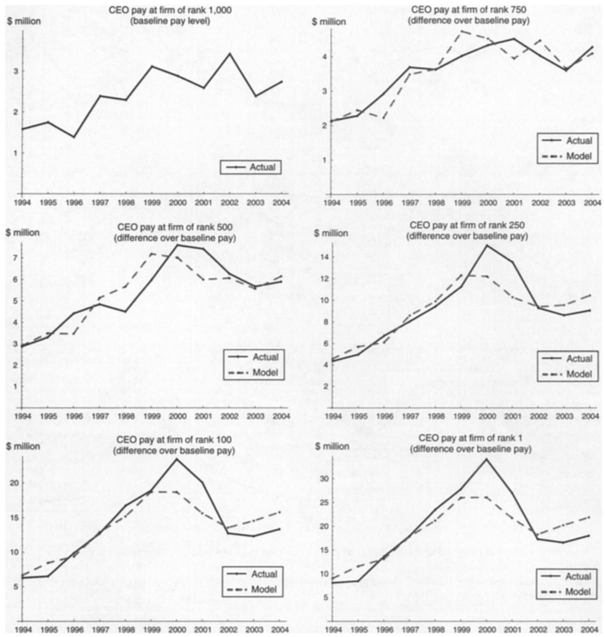

FIGURE 6. THE DIFFERENCE IN PAY BETWEEN THE CEOS OF SELECTED RANKS AND THE BASELINE (1,000TH) CEO

Notes: "Actual" pay refers to the smoothed CEO pay and "Modeled" refers to the pay level generated by the assignment model while imposing time-invariant distributions of ability and firm size and size-neutral productivity growth. All values are in 2004 dollars. Parameter scenarios make only a negligible (below 1 percent) difference to the fitted wages so only one fit (scenario A from Table 1) is shown.

Figure 6: Pay differences under time-invariant calibration

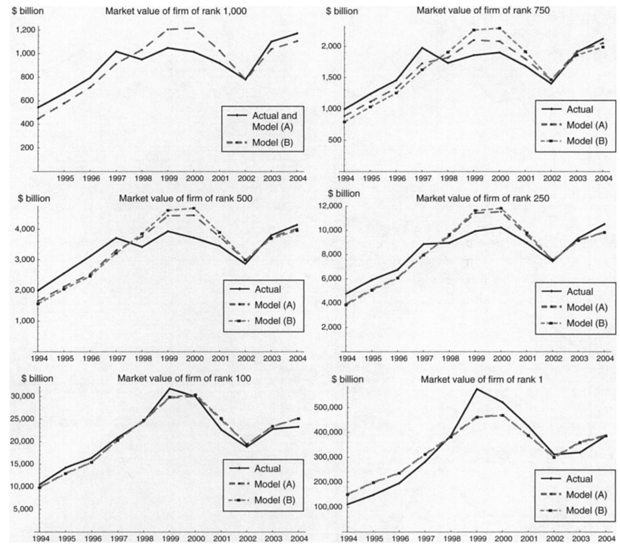

FIGURE 7. PREDICTED MARKET VALUES IN THE TIME-INVARIANT CALIBRATION

Notes: All values are in 2004 dollars. The market values are calculated under the two parameter scenarios that generate the most extreme results (see Table 1).

Figure 7: Market values under time-invariant calibration

9 Short summary

- The paper builds a two-sided assignment model of CEOs and firms.
- CEOs differ in ability; firms differ in exogenous natural scale.
- Complementarity between CEO ability and firm scale generates positive assortative matching.
- Equilibrium CEO pay reflects differential rents from scarce talent and scarce large-firm positions.
- Observed CEO pay and market values are used to infer latent ability and firm-quality profiles.
- The inferred ability differences among top CEOs are small.
- Most CEO pay differences are due to firm heterogeneity, not huge differences in CEO ability.
- Replacing all top 1,000 CEOs with the best observed CEO type would raise total surplus by only a few billion dollars.
- The high pay of top CEOs can be consistent with competitive markets, but that does not mean marginal CEO talent creates huge social value.
- The paper separates the competitive explanation of CEO pay from the economic value of CEO ability.

10 Conclusion

10.1 Core conclusion

Terviö shows that a competitive assignment model can rationalize very high CEO pay while implying relatively small differences in managerial ability. The joint distribution of CEO compensation and firm market value suggests that firm-specific scale differences explain most of the variation in CEO pay.

10.2 Economic intuition

Small differences in CEO ability can be magnified when they are matched to very large firms. Large firms pay more not necessarily because their CEOs are vastly more talented, but because any talent difference is more valuable at larger scale. At the same time, firms themselves differ in natural scale, and these differences generate large Ricardian rents that show up in CEO pay and firm values.

10.3 Incremental contribution

The paper's main contribution is to use a two-sided assignment model to distinguish CEO ability from firm heterogeneity. It demonstrates that competitive matching can explain high CEO pay while also showing that the social value of scarce CEO talent is modest relative to firm scale. This distinction deepens the executive-compensation debate by separating *competitive pay determination* from *the economic importance of ability*.

10.4 Takeaway

The assignment approach changes the interpretation of CEO pay. High compensation is not direct evidence that CEOs create enormous additional value, nor is it automatically evidence of market failure. It is the equilibrium price of scarce managerial talent matched with scarce large-firm opportunities.